(ii) **The commutative property** For any two complex numbers z_1 and z_2 $z_1 + z_2 = z_2 + z_1$.	(ii) **The commutative property** For any two complex numbers z_1 and z_2 $z_1 z_2 = z_2 z_1$.
(iii) **The associative property** For any three complex numbers z_1, z_2, and z_3 $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$.	(iii) **The associative property** For any three complex numbers z_1, z_2, and z_3 $(z_1 z_2) z_3 = z_1 (z_2 z_3)$.
(iv) **The additive identity** There exists a complex number $0 = 0 + 0i$ such that, for every complex number z, $z + 0 = 0 + z = z$ The complex number $0 = 0 + 0i$ is known as additive identity.	(iv) **The multiplicative identity** There exists a complex number $1 = 1+0i$ such that, for every complex number z, $z1 = 1z = z$ The complex number $1 = 1 + 0i$ is known as multiplicative identity.
(v) **The additive inverse** For every complex number z there exists a complex number $-z$ such that, $z + (-z) = (-z) + z = 0.$ $-z$ is called the additive inverse of z.	(v) **The multiplicative inverse** For any nonzero complex number z, there exists a complex number w such that, $z\,w = w\,z = 1.$ w is called the multiplicative inverse of z. w is denoted by z^{-1}.
(vi) **Distributive property (multiplication distributes over addition)** For any three complex numbers z_1, z_2, and z_3 $z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$ and $(z_1 + z_2) z_3 = z_1 z_3 + z_2 z_3$.	

Let us now prove some of the properties.

Property

The commutative property under addition

For any two complex numbers z_1 and z_2, we have $z_1 + z_2 = z_2 + z_1$.

Proof

Let $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$, and x_1, x_2, y_1, and $y_2 \in \mathbb{R}$,

$$\begin{aligned} z_1 + z_2 &= (x_1 + iy_1) + (x_2 + iy_2) \\ &= (x_1 + x_2) + i(y_1 + y_2) \\ &= (x_2 + x_1) + i(y_2 + y_1) \qquad (\text{since } x_1, x_2, y_1, \text{ and } y_2 \in \mathbb{R}) \\ &= (x_2 + iy_2) + (x_1 + iy_1) \\ &= z_2 + z_1. \end{aligned}$$

Property ■

Inverse Property under multiplication

Prove that the multiplicative inverse of a nonzero complex number $z = x + iy$ is $\frac{x}{x^2 + y^2} + i\frac{-y}{x^2 + y^2}$.

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